

Failure Case F-004: Silent Redesign Failure

Severity: HIGH | Version: v0.1-prototype | Date: 2026-04-14

Trigger	Level Designer redesign produced invalid JSON during a debug cycle.
What happened	<code>_redesign_levels()</code> caught all exceptions with: <code>except Exception: new_levels = []</code> The redesigned levels were silently discarded. The broken level was never replaced, so the pipeline ran all 3 debug cycles doing nothing useful -- the same unsolvable level was re-tested each cycle.
Pipeline behavior	Cycle 1: QA fails Level 5 -> Dev routes to Level Designer -> LLM returns invalid JSON -> <code>except Exception: new_levels = []</code> -> QA re-tests: Level 5 still unsolvable Cycle 2: Same as Cycle 1. No progress. Cycle 3: Same as Cycle 1. No progress. Result: 3 wasted debug cycles. The feedback loop appeared to work but did nothing.
Root cause	Overly broad exception handling that prioritized graceful degradation over correctness. The silent failure broke the feedback loop -- QA kept failing, Developer kept routing, but Level Designer's fix was being thrown away every time.
Fix applied	Added retry logic with error feedback in <code>_redesign_levels()</code> . On parse failure, a second LLM call is made with the specific error message. Also improved the redesign prompt to include original level dimensions and solver-timeout guidance.
Governance lesson	Silent failure in a feedback loop is worse than a crash. The system appeared to be working (3 cycles completed) but was doing nothing. Observability and explicit error handling are essential for multi-agent reliability.